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What is the difference between a version like `4.0.2` and another like `4.x` and yet another like `4.0.x` in the context of semantic versioning for Node.js dependencies?

In the context of semantic versioning, especially for Node.js dependencies, the version numbers like `4.0.2`, `4.x`, and `4.0.x` represent different approaches to specifying dependency versions. Here's how they differ:

- 1. `4.0.2`:** This is a specific version number. It means that the dependency will be exactly at version 4.0.2. When you specify a dependency in this way, you are indicating that your application should use that particular version and no other. This is often used when you know that your application is compatible with that specific version and you want to avoid any unexpected changes that might come with newer versions.
- 2. `4.x`:** This version number uses a wildcard for the minor version. It means that you are okay with using any version that starts with `4.` as long as the major version (4 in this case) does not change. For example, it can install 4.1.0, 4.2.3, etc., but it will not upgrade to 5.0.0. This approach is more flexible than specifying a specific version, allowing for minor updates that typically include new features and backwards-compatible changes.
- 3. `4.0.x`:** Here, the wildcard is used for the patch version. This means you are comfortable using any patch version of 4.0.x. For example, it could install 4.0.3, 4.0.4, etc., but it will not upgrade to 4.1.0. Patch versions are usually for bug fixes and other minor changes that are backwards-compatible. This is more restrictive than `4.x` but more flexible than specifying a specific version like `4.0.2`.

In summary:

- `4.0.2` locks the dependency to an exact version.
- `4.x` allows updates within the major version 4, including minor and patch updates.
- `4.0.x` allows updates within the minor version 4.0, including only patch updates.

This approach to versioning helps maintain a balance between the stability of your application and the benefits of receiving bug fixes and new features.

